Radical radiotherapy for bladder cancer

Radical radiotherapy means using high doses of radiotherapy to try to cure bladder cancer.

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What is radical radiotherapy for bladder cancer?

Radiotherapy uses high-energy rays called radiation to treat cancer. It destroys cancer cells in the area where the radiotherapy is given while doing as little harm as possible to normal cells. Radical radiotherapy means using high doses of this treatment to try to cure cancer.

This treatment is used to treat muscle-invasive or locally advanced bladder cancer. It is given with chemotherapy or a treatment called carbogen and nicotinamide. You may be offered it instead of surgery to remove the bladder. This is called a cystectomy.

We have more information about making decisions about bladder cancer treatment.

Palliative radiotherapy for bladder cancer

Radiotherapy may also be given if advanced bladder cancer is causing symptoms such as pain or bleeding. This is sometimes called palliative radiotherapy.

Palliative radiotherapy aims to improve the symptoms. It is a shorter course of treatment than radical radiotherapy.

For bladder symptoms you usually have 3 to 5 treatments over a week. Sometimes, you will have 1 single treatment.

For bone pain symptoms you may only need 1 treatment. Or you may need up to 5 treatments.

We have more information about treating advanced bladder cancer.

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Before radical radiotherapy

Your cancer doctor or nurse will explain what your treatment involves and possible side effects. They will explain if you need other treatments before you have the radiotherapy, such as:

- a small operation to remove as much cancer as possible from inside the bladder – called a transurethral resection of a bladder tumour or TURBT
- chemotherapy to shrink the cancer and reduce the risk of it coming back.

Radiotherapy treatment may affect your fertility. Fertility is the ability to get pregnant or make someone pregnant. If you are worried about this, it is important to talk to your doctor before your treatment starts.

How radical radiotherapy is given

You have radiotherapy in the radiotherapy department at the hospital. Treatment is given as a series of short, daily treatments. It is usually given on weekdays (Monday to Friday).

You may have to travel to a hospital further away than your local hospital to have treatment.

Planning your radiotherapy treatment

Radiotherapy is always carefully planned by a team of experts. They will plan your treatment so it does as little harm as possible to normal cells. The radiotherapy will be aimed precisely at the cancer.

On your first visit to the radiotherapy department, you will usually have a CT scan of the area that is being treated. The information from the scan is sent to a planning computer.

Your radiotherapy team use information from this scan to plan:

- the dose of radiotherapy
- the precise area to be treated.

During your scan, you lie still on a narrow bed. This is sometimes called a table. If you feel uncomfortable, tell the radiographers so they can help. It is important that you are comfortable, because you need to be able to lie in the same position for all your radiotherapy treatment.

You may have some small, permanent markings made on your skin. The marks are about the size of a pinpoint. They are made in the same way as a tattoo. The marks help the radiographer make sure you are in the correct position for each session of radiotherapy.

These marks will only be made with your permission. If you are worried about them or already have a tattoo in the treatment area, tell your radiographer. They can discuss this with you.

The radiographer uses laser lights to position you for your radiotherapy treatment. These are not harmful to your skin.

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How long does radiotherapy take?

You usually have radical radiotherapy for bladder cancer as an outpatient. Each treatment takes 10 to 15 minutes.

How often you have treatment may depend on your general fitness and other health conditions. The treatments are usually given once a day from Monday to Friday, with a rest at the weekend, for 4 to 7 weeks.

If you are less fit, you may have 1 treatment a week for 6 weeks. Your radiotherapy team will explain how many treatments you will have and when you will have them.

Having radiotherapy

At the beginning of each session of radiotherapy, the radiographer will make sure you are in the correct position and that you are comfortable. When everything is ready, they leave the room and the treatment starts.

The radiographers can see and hear you from outside the room. There is usually an intercom, so you can talk to them if you need to during your treatment.

Radiotherapy is not painful, but you will have to lie still for a few minutes during the treatment.

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Having other treatment with radiotherapy

You may have other treatment in combination with radical radiotherapy. This is to help treatment work better. You may have radiotherapy with:

Chemotherapy

Chemotherapy given with radiotherapy is called chemoradiation. You may have chemotherapy every weekday in week 1 and week 4 of your radiotherapy. Or, you may have chemotherapy once a week during the radiotherapy course. Chemoradiation may cause more severe side effects, so you need to be well enough to have this.

Carbogen and nicotinamide

You may have a gas called carbogen that you breathe in for a few minutes before and during your radiotherapy. You also take nicotinamide tablets.

Carbogen and nicotinamide help increase the amount of oxygen that the cancer cells get. Lack of oxygen protects the cancer cells from radiotherapy damage. They are more likely to be destroyed by radiotherapy if they have lots of oxygen. This treatment is not available in all hospitals. Your doctor will tell you if this is an option for you.

Side effects of radiotherapy to the bladder

Your doctor, nurse or radiographer will talk about side effects with you. Tell them about any side effects you have during or after treatment. There are often things that can help.

Side effects may build up towards the end of your course of radiotherapy. After treatment finishes, side effects usually get better over a few weeks or months.

Radiotherapy to the bladder and nearby areas can cause the following side effects.

Bowel effects

These may include:

- loose stools (poo) or diarrhoea
- cramping pains in your tummy (abdomen) or back passage (rectum)
- feeling you need to go to the toilet although your bowel is empty (tenesmus)
- passing mucus or blood when you empty your bowels
- sore skin around the back passage.

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Your specialist may give you anti-diarrhoea drugs to help. Your nurse or radiographer may suggest making changes to your diet, such as reducing how much fibre you eat. It is important to drink at least 2 litres (3½ pints) of fluids a day.

Bladder effects

These include:

- passing urine (peeing) more often than usual
- passing urine during the night
- a burning feeling when you pass urine (similar to a urine infection)
- a poor flow of urine
- a feeling that you are not able to wait when you need to pass urine
- leaking small amounts of urine (incontinence)
- blood in your urine.

Your doctor can give you medicines to help. They may also ask you for a urine sample to check if you have an infection. An infection is a common cause of symptoms and is easy to treat.

Drinking fluids will help. Try to drink around 2 litres (3½ pints) a day. Try to drink water instead of drinks that may irritate the bladder. These include:

- drinks containing caffeine, such as tea, coffee, drinking chocolate and cola
- alcohol
- fizzy drinks
- acidic drinks – for example, orange or grapefruit juice
- drinks with artificial sweeteners – for example, diet or ‘light’ drinks.

We have more information about tips for coping with bladder or bowel changes.

Effects on the skin

The skin in the area being treated may become red and sore or itchy. It may also become darker. Tell your radiotherapy team if you notice any skin changes. They can give you advice about looking after your skin.

Tiredness

Radiotherapy often makes people feel tired. Tiredness (fatigue) may get worse as treatment goes on. If you are having radiotherapy alongside other treatments, such as surgery or chemotherapy, you may feel more tired. But there are things you can do to help, such as:

- get plenty of rest
- do some gentle exercise, such as short walks
- eat a healthy diet and drink plenty of fluids
- ask others for help with everyday jobs.

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After treatment finishes, you may continue to feel tired for weeks or months. If it does not get better, tell your cancer doctor or specialist nurse.

We have more information about coping with tiredness (fatigue).

Hair loss

Radiotherapy only affects hair in the area treated. You may lose some of your pubic hair. It usually grows back after treatment finishes. It may take several months to grow back, although it depends on the dose of radiotherapy you have. Sometimes, hair loss is permanent.

If you have chemotherapy with radiotherapy, the chemotherapy drugs may cause hair loss in other areas of the body, such as from your head. Your doctor or nurse will explain what is likely to happen. Some chemotherapy drugs cause all or most of your hair to fall out.

We have more information about hair loss.

Possible late effects of radiotherapy to the bladder

Some people may have side effects that do not improve, or side effects that happen months or years after treatment. These are called long-term or late effects. Your doctor or nurse will explain these to you.

Late effects of radiotherapy to the bladder may include:

Bowel or bladder changes

If your bowel is affected, you may have ongoing loose stools (poo) or diarrhoea. If your bladder is affected, it may not hold as much urine (pee) so you may need to pass urine more often.

The blood vessels in the bowel and bladder can become more fragile. If this happens, you may see blood in your urine or stools. If you notice blood, tell your doctor straight away. They will do tests and give you the right treatment.

Always tell your cancer doctor, nurse or GP about any bowel or bladder symptoms you have. They can give you advice and may do some tests. Often symptoms can be treated.

Vaginal changes

The vagina may become narrower and less stretchy after pelvic radiotherapy. This can make it more uncomfortable to have internal examinations. It can also make penetrative sex difficult or uncomfortable.

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A vaginal lubricant or moisturiser may help. Sometimes using a vaginal dilator will help. Your cancer doctor or nurse can give advice about this.

Menopause

Radiotherapy is likely to cause menopause if you have not already been through it. This means monthly periods will stop and it is no longer possible to get pregnant.

Your doctor or nurse can also give you information about managing menopausal symptoms and your options for possible fertility treatments to start a pregnancy in the future.

Erection problems (erectile dysfunction)

Radiotherapy can cause problems getting or keeping an erection. There are treatments that can help with erection problems.

Fertility problems

Radiotherapy to the pelvic area can affect your ability to get pregnant or make someone pregnant and may cause permanent infertility. If you are worried about this, it is important to talk to your doctor before you start radiotherapy.

There are different ways late effects can be treated or managed.

We have more information about managing bladder late effects and managing bowel late effects.

Getting support

Macmillan is also here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on 0808 808 00 00.
- Chat to our specialists online.
- Visit our bladder cancer forum to talk with people who have been affected by bladder cancer, share your experience, and ask questions.

About our information

References

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Below is a sample of the sources used in our bladder cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Mottet N, Bellmunt J, Briers E, et al. Non-muscle-invasive bladder cancer (TaT1 and CIS). European Association of Urology (Internet), 2021. Available from uroweb.org/guideline/non-muscle-invasive-bladder-cancer (accessed September 2021).

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Reviewers

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, Dr Ursula McGovern, Consultant Medical Oncologist.

Our cancer information has been awarded the PIF TICK. Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

Reviewed: 01 November 2022 | Next review: 01 November 2025

This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

How we can help

Macmillan Support Line

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The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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